Healthcare Data Pipeline Project Documentation

1. Project Overview:

Healthcare organizations generate large amounts of data such as patient records, admissions, and billing information. This data is often unstructured and difficult to analyze.

This project focuses on building a **scalable data pipeline** that collects, cleans, and transforms raw healthcare data into meaningful insights for decision-making.

2. Objectives:

- Centralize healthcare data
- Improve data quality and consistency
- Enable easy analysis using structured data
- Build a scalable and reliable system
- Prepare data for dashboards and machine learning

3. Solution Approach:

We implemented a **Medallion Architecture** with three layers:

- **Bronze (Raw Data)**
- **Silver (Clean Data)**
- **Gold (Business Insights)**

This ensures a structured flow from raw data to useful insights.

4. System Architecture:

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Data Source → AWS S3(any other) → Bronze → Silver → Gold → Analytics
```

5. Bronze Layer (Raw Data):

- Data is ingested from AWS S3(adls, databricks catalog etc)
- Stored as-is without modification
- Maintains original data for traceability

👉 *Purpose: Preserve raw data*

6. Silver Layer (Data Cleaning & Transformation):

- Removes null values and duplicates
- Standardizes formats (dates, columns)
- Applies data transformations
- Implements **SCD Type 2 using MERGE** for tracking changes

👉 *Purpose: Ensure clean and reliable data*

7. Gold Layer (Business Insights):

- Performs aggregations and analysis
- Creates multiple tables for different business needs

Examples:

- Patient count per hospital
- Hospital ranking
- Contribution by medical condition

👉 *Purpose: Provide ready-to-use insights*

8. Technology Stack:

- **Python & PySpark** → Data processing
- **SQL (Spark SQL)** → Analytical queries
- **Databricks + Delta Lake** → Data pipeline & storage
- **AWS S3, ADLS etc** → Data ingestion
- **Delta Live Tables (DLT)** → Pipeline automation
- **IAM (Conceptual)** → Secure access

9. Data Engineering Concepts:

- **ETL Process**
  - Extract → S3, ADLS etc
  - Transform → Silver
  - Load → Gold
- **Big Data Processing** → Using PySpark
- **Medallion Architecture** → Layered design
- **SCD Type 2 + MERGE** → Tracks historical data changes

10. Processing Approach:

- Current system uses **batch processing**
- Can be extended to **real-time using Kafka**

11. Future Scope: BI Dashboard:

Overview:

The Gold layer data can be used to create dashboards for visualization.

Implementation:

- Tools: Databricks SQL Dashboard / Power BI / Tableau

- Data Source: Gold tables

Key Insights:

- Total patients per hospital
- Average billing
- Hospital rankings
- Condition-based analysis

Benefits:

- Easy-to-understand visuals
- Faster decision-making
- Real-time analytics

12. Future Scope: MLOps:

Overview:

MLOps enable the use of data for predictive analytics.

Use Cases:

- Predict high-risk patients
- Forecast hospital workload
- Estimate billing trends

Process:

1. Train models using gold data
2. Deploy models for predictions
3. Monitor performance
4. Retrain when needed

Benefits:

- Data-driven predictions

- Automated model lifecycle
- Improved accuracy over time

13. Conclusion:

This project demonstrates how raw healthcare data can be transformed into meaningful insights using a scalable data pipeline. By adding BI dashboards and MLOps, the system can evolve into a complete data-driven solution that supports both analysis and prediction.